RESUME-LENS: INTEGRATIVE TASK 1

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Course: Computing and Discrete Structures III
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GUIDING QUESTION

Can formal language models automatically check whether a résumé matches a professional profile, even though each person describes their skills differently?

WHAT IS RESUMELENS?

Each person lists their skills differently (e.g., JS, Javascript, JavaScript), and each position requires a unique combination. Reviewing this manually is slow and inconsistent. ResumeLens reads a resume and checks whether the listed skills match each profile's pattern using formal language theory—regular expressions, transducers, automata, and grammars—yielding one of two outcomes per profile (accepted or rejected). The tool does not rank candidates or make hiring decisions; it only verifies whether the text conforms to a defined pattern.

PROJECT ARCHITECTURE



Pipeline.py (system orchestrator)

PROFILES

The profiles proposed for consideration are as follows:

Full Stack Developer: Front-End y Backend.
- JavaScript or TypeScript
- React, Angular, or Vue
- Node.js, Django, Spring Boot, or a similar backend technology
- SQL or NoSQL databases
- REST APIs
- Git

Machine Learning Engineer : Self-learning systems that learn from data.
- Python
- Pandas or NumPy
- Scikit-learn
- TensorFlow, or PyTorch
- Machine-learning model development
- SQL
- Git

Data Scientist: Data analysis.
- Python or R
- Pandas or NumPy
- Scikit-learn
- Matplotlib or Seaborn
- Statistics and probability
- SQL
- Git

Software Architect: Designs the system structure.
- Java, C#, or Python
- Software design patterns
- Microservices
- Docker or Kubernetes
- AWS, Azure, or GCP
- REST APIs or GraphQL
- SQL or NoSQL databases
- Git

LITERATURE REVIEW

SOURCE 1: RESUMATCHER: A PERSONALIZED RESUME-JOB MATCHING SYSTEM

A system that matches resumes with job offers. Limitations: Its output is a similarity measure for recommending jobs, not an accepted/rejected response against a pattern.

Guo, S., Alamudun, F., & Hammond, T. (2016). RésuMatcher: A personalized résumé-job matching system. Expert Systems With Applications, 60, 169–182. https://doi.org/10.1016/j.eswa.2016.04.013

SOURCE 2: TEXT NORMALIZATION FOR LOW-RESOURCE LANGUAGES OF AFRICA

They study the effect of text normalization for low-resource African languages using a normalizer based on transducers in Python. Limitations: **Normalizing the text is useful,** but it doesn't say whether transducers are better than other methods.

Zupon, A., Crew, E., & Ritchie, S. (2021). Text normalization for low-resource languages of Africa. En arXiv [cs.CL]. https://doi.org/10.48550/arXiv.2103.15845

SOURCE 3: WEB-BASED RECRUITING

They analyze and structure French CVs using GATE rules to convert them into the Europass format. Limitations: **The study shows an improvement in extraction and does not mention** classification results.

Amdouni, S., & Ben abdessalem Karaa, W. (2010). Web-based recruiting. ACS/IEEE International Conference on Computer Systems and Applications – AICCSA 2010, 1–7. https://ieeexplore.ieee.org/abstract/document/5587018

Based on the aforementioned background, my team made the following decisions: to use regular expressions for extraction and add automata to validate whether a candidate meets each profile, while verifying formal patterns (accepted/rejected) rather than scoring or ranking candidates. Transducers were chosen because the variants are few and well-known; the 'pyformlang' library will be used for normalization via transducers, eliminating the need for training data.

SYSTEM PROCESS

STAGE 1: FINDING THE INFORMATION

A regular expression is like a "search rule" that describes what a type of text looks like.

What does each one recognize? Almost all of them are a finite combination of words. For example, the one for years of experience recognizes digits followed by " years of experience," and the email one recognizes text + @ + domain + .TLD (Top-Level Domain). A TLD is the final extension of an address (such as .com, .org, or .mx).

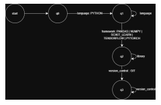
STAGE 2: STANDARDIZE THE NAMES

Cada transductor es M = (Q, Σ, Γ, δ, ω, q0, F):
Q = {q0, qf}: estado inicial y estado final.
Σ (entrada): las variantes de escritura.
Γ (salida): los nombres oficiales.
δ: por cada variante hay una transición q0 --> qf.
ω: la salida de esa transición, o sea el nombre oficial.
q0 es el estado inicial y F = {qf}.



STAGE 3: DECIDE WHETHER THE PROFILE MEETS THE REQUIREMENTS.

Cada autómata es M = (Q, Σ, δ, q0, F). Para Machine Learning Engineer:
Q = {q0, q1, q2, q3}
Σ = {PYTHON, PANDAS, NUMPY, SCIKIT_LEARN, TENSORFLOW, PYTORCH, GIT}
δ: q0 – PYTHON --> q1; q1 – PYTHON --> q1; q1 – (librería) --> q2; q2 – (librería) --> q2; q2 – GIT --> q3; q3 – GIT --> q3. Lo que no está definido se rechaza.
q0 es el estado inicial y F = {q3}.



STEP 4: DESCRIBE AND SHOW THE RESULT

A grammar defines the rules for how something is written. We created a small language (a DSL) to describe the candidate. If it is valid, an HTML page is generated. If the text violates a rule, it is rejected with an error.



SYSTEM RESULTS

Juan_Jaramillo Resume Process

STAGE	RESULT
Detection	JS, React.js, NodeJS, Postgres, Git
Transformation	JAVASCRIPT, REACT, NODE_JS, POSTGRESQL, GIT (already ordered)
Classification	FULL_STACK_DEVELOPER: ACCEPTED; the other three, REJECTED
Validation	A profile with MACHINE_LEARNING_ENGINEER = MAYBE was rejected by the grammar.

A single person can fit more than one profile. For example, someone who fits the Data Scientist profile also fits the Machine Learning Engineer profile, because they share skills.

SYSTEM DECISIONS

The transducers operate using complete words, since our variants are known, making each rule simple. Each profile requires its core components (language, main library, and Git) and is always ordered consistently prior to classification, ensuring the result does not depend on the order in which the resume was written. This entails one deterministic automaton per profile—which is easy to test—and a single program for all four profiles.



LIMITATIONS AND POTENTIAL

ResumeLens recognizes only the skills on our list and reviews the essentials of each profile, meaning similar profiles may be accepted simultaneously. Its language processing handles only a single work experience, and the outcome is binary—accepted or rejected—without specifying the reason. Planned improvements include adding more writing and profile variations, supporting multiple work experiences and educational backgrounds, and indicating what was missing when a candidate is rejected.

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